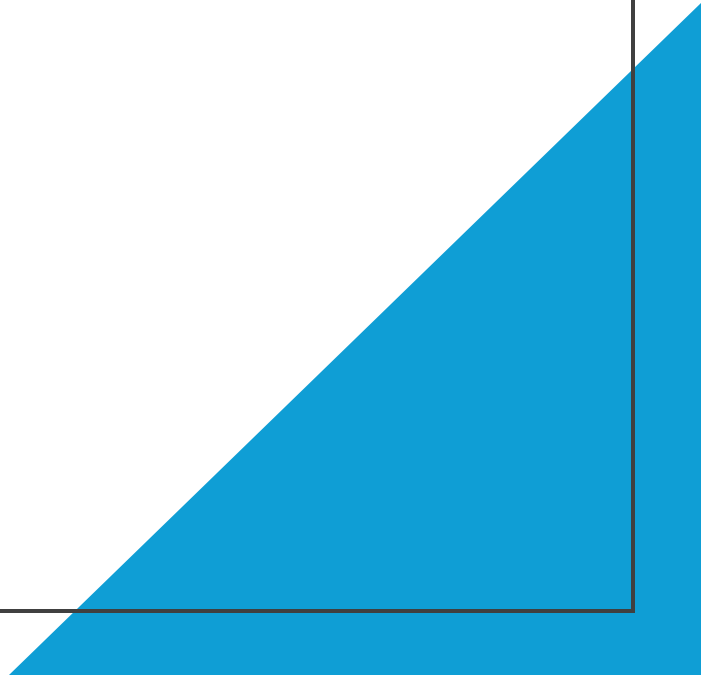


SmartHome Sistemas Inteligentes

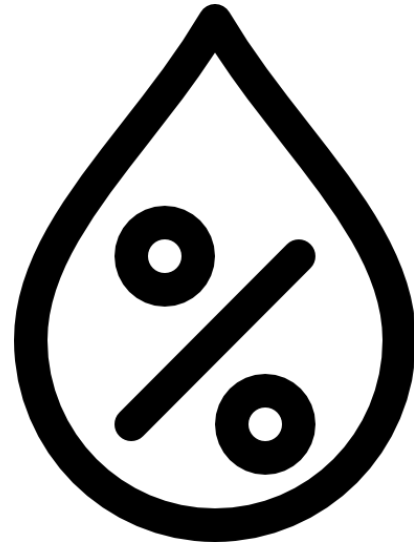
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- Ángel Prieto Méndez
- Daniel Peláez López
- Javier Ferrer Gatell
- Adrián Calvo Moyano



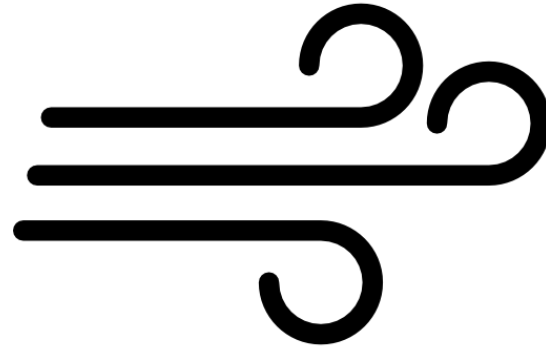
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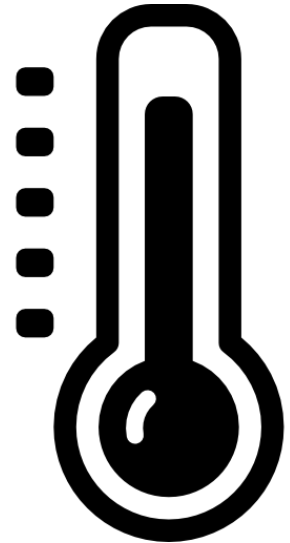
Lluvia



Humedad



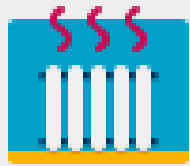
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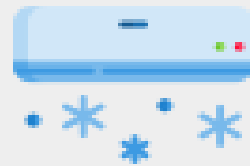
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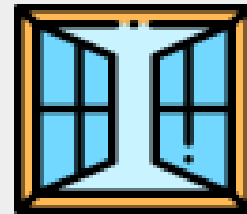
Estado de los dispositivos



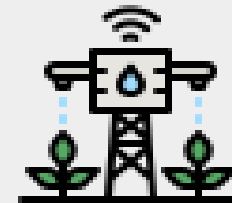
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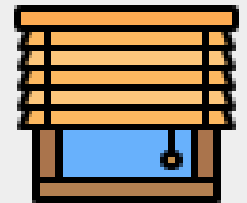
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ON



OFF



ON

Historial de acciones

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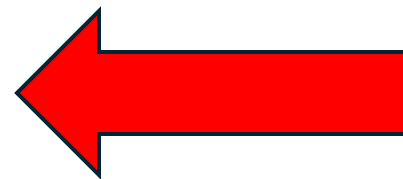
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Humedad: 79.6%

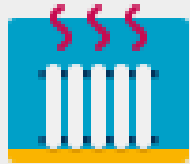
Velocidad del viento: 17.7 Km/h

Lluvia: no

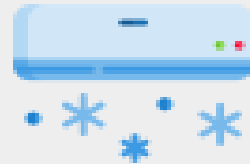




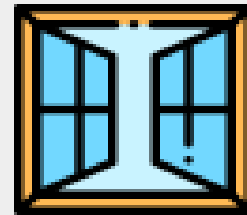
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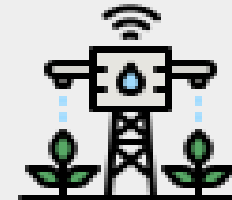
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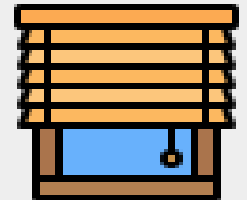
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ON



OFF



ON

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Datos recibidos

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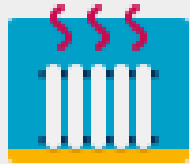
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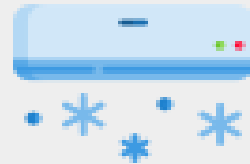
Lluvia: no



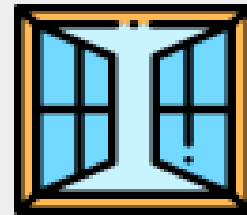
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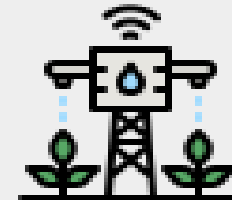
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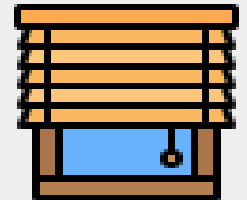
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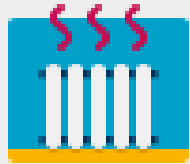
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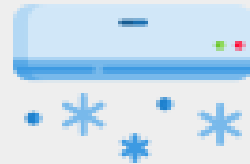
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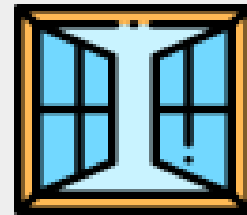
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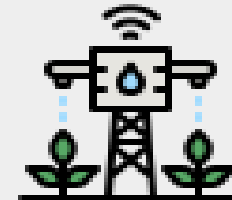
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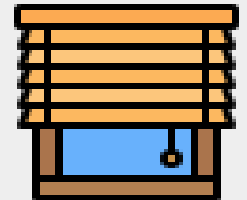
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ON

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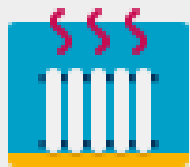
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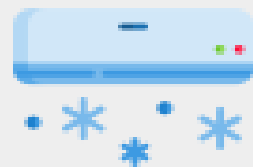
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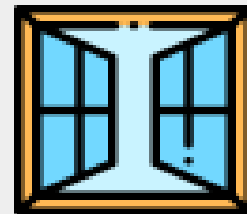
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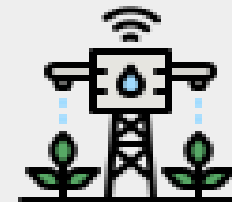
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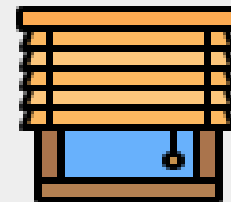
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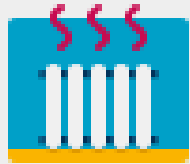
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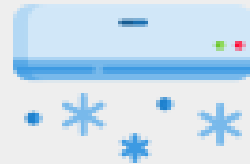
Temperatura: 23.0°C
Humedad: 79.6%
Velocidad del viento: 17.7 Km/h
Lluvia: no



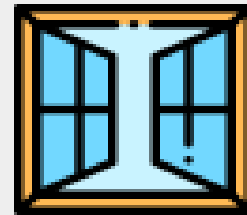
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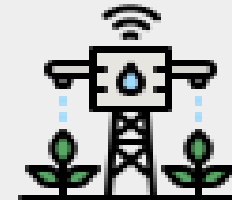
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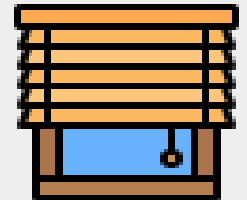
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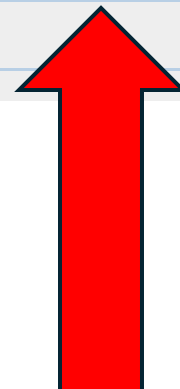
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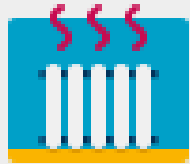
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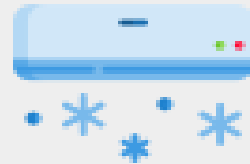




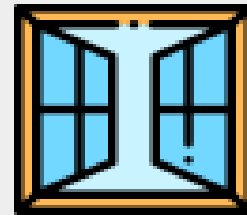
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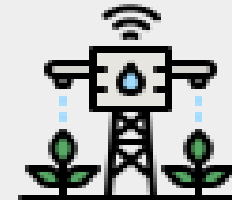
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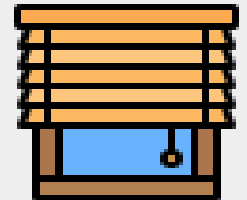
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OFF



ON

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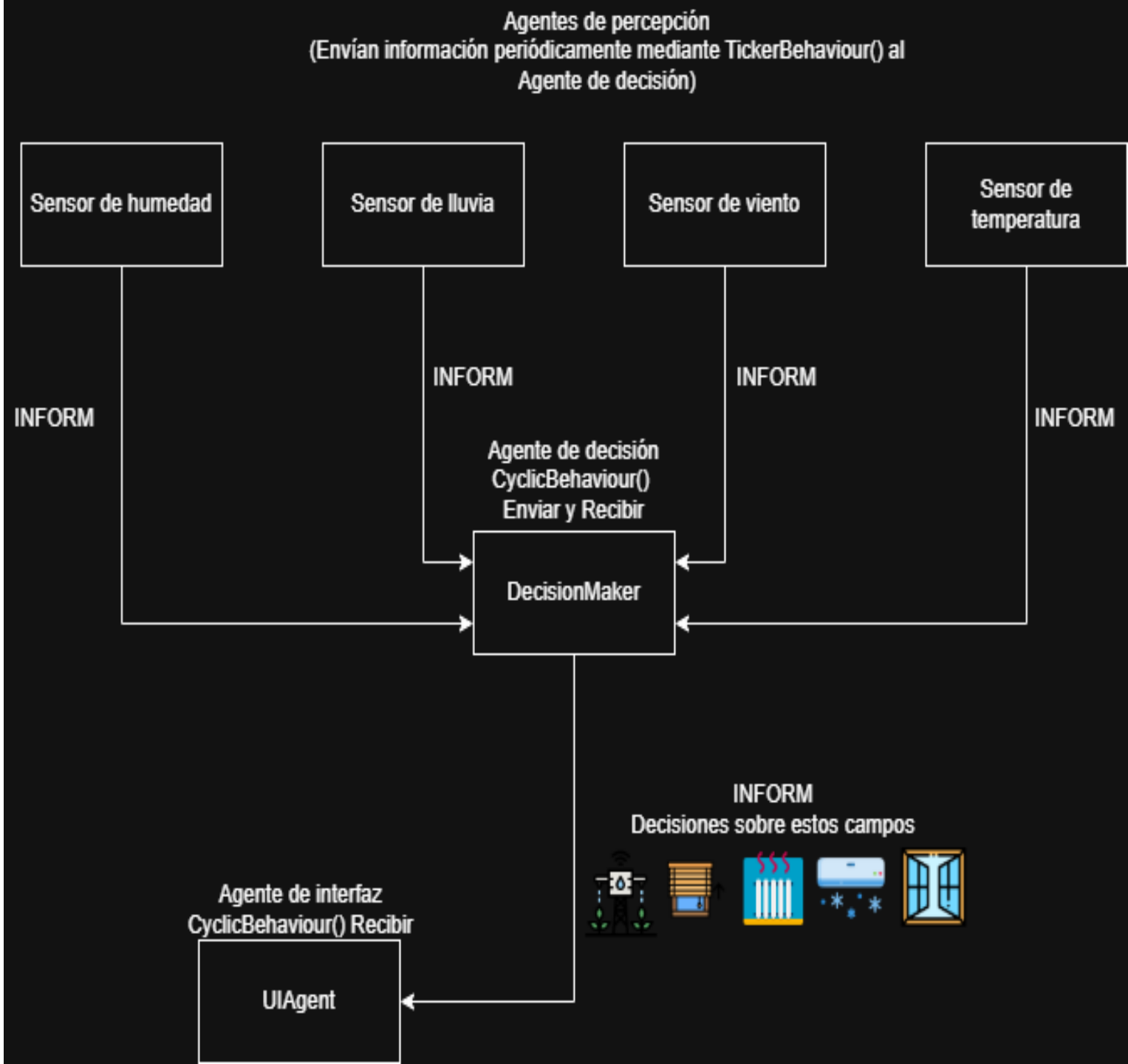
Temperatura: 23.0°C

Humedad: 79.6%

Velocidad del viento: 17.7 Km/h

Lluvia: no

Diseño de la arquitectura



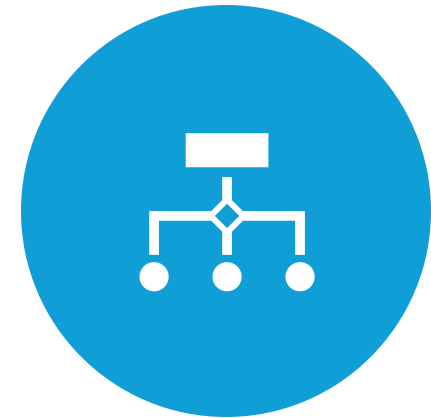
Ontología



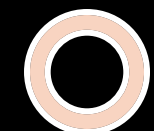
USAREMOS DOS ONTOLOGÍAS DISTINTAS PARA
EL ENVÍO Y RECEPCIÓN DE MENSAJES ACL.



ENTRE LOS AGENTES DE PERCEPCIÓN Y EL
AGENTE DE DECISIÓN, USAREMOS LA
ONTOLOGÍA "WEATHER-ONTOLOGY".



ENTRE EL AGENTE DE DECISIÓN Y EL AGENTE
DE LA INTERFAZ GRÁFICA USAREMOS LA
ONTOLOGÍA "ACTION-ONTOLOGY".



Clases auxiliares

Agent Model

```
1 package es.upm.smarthome;
2
3 public enum AgentModel {
4     TEMPERATURE_SENSOR("TemperatureSensor"),
5     HUMIDITY_SENSOR("HumiditySensor"),
6     WIND_SENSOR("WindSensor"),
7     RAIN_SENSOR("RainSensor"),
8     DECISION("Decision"),
9     UI("UI"),
10    DESCONOCIDO("Desconocido");
11
12    private final String value;
13
14    AgentModel(String value) { this.value = value; }
15
16    public String getValue() { return value; }
17
18    public static AgentModel getEnum(String value) {
19        switch (value) {
20            case "TemperatureSensor": return TEMPERATURE_SENSOR;
21            case "HumiditySensor": return HUMIDITY_SENSOR;
22            case "WindSensor": return WIND_SENSOR;
23            case "RainSensor": return RAIN_SENSOR;
24            case "Decision": return DECISION;
25            case "UI": return UI;
26            default: return DESCONOCIDO;
27        }
28    }
29 }
```

Main

```
public class Main {
    private static AgentContainer cc;

    private static void loadBoot() {
        jade.core.Runtime rt = jade.core.Runtime.instance();
        rt.setCloseVM(true);
        System.out.println("Runtime created");

        Profile profile = new ProfileImpl(null, 1200, null);
        System.out.println("Profile created");

        cc = rt.createMainContainer(profile);
        System.out.println("Main container created");

        try {
            // Lanzar RMA (interfaz gráfica de JADE)
            cc.createNewAgent("rma", "jade.tools.rma.rma", new Object[0]).start();

            // Crear agentes del sistema
            cc.createNewAgent("TempSensor", TemperatureSensorAgent.class.getName(), new Object[]{}).start();
            cc.createNewAgent("HumiditySensor", HumiditySensorAgent.class.getName(), new Object[]{}).start();
            cc.createNewAgent("WindSensor", WindSensorAgent.class.getName(), new Object[]{}).start();
            cc.createNewAgent("RainSensor", RainSensorAgent.class.getName(), new Object[]{}).start();
            cc.createNewAgent("DecisionMaker", DecisionAgent.class.getName(), new Object[]{}).start();
            cc.createNewAgent("UI", UIAgent.class.getName(), new Object[]{}).start();

            System.out.println("All agents launched");
        } catch (StaleProxyException e) {
            System.err.println("Error launching agents");
            e.printStackTrace();
            System.exit(1);
        }
    }

    public static void main(String[] args) {
        System.out.println("Starting Smart Home MAS...");
        loadBoot();
        System.out.println("System ready");
    }
}
```

Agent Base

```
public abstract class AgentBase extends Agent {
    protected AgentModel type;
    protected String[] params;

    protected void setup() {
        super.setup();
        if (getArguments() != null) {
            this.params = Arrays.asList(getArguments()).toArray(new String[getArguments().length]);
        }
    }

    public DFAgentDescription[] getAgentsDF(AgentModel type) {
        DFAgentDescription template = new DFAgentDescription();
        ServiceDescription templateSd = new ServiceDescription();
        templateSd.setType(type.getValue());
        template.addServices(templateSd);
        DFAgentDescription[] result = new DFAgentDescription[0];
        try {
            result = DFService.search(this, template);
        } catch (FIPAException e) {
            e.printStackTrace();
            loge("DFService.search failed");
        }
        return result;
    }

    public void registerAgentDF() {
        DFAgentDescription dfd = new DFAgentDescription();
        dfd.setName(getAID());
        ServiceDescription sd = new ServiceDescription();
        sd.setType(this.type.getValue());
        sd.setName(getLocalName());
        dfd.addServices(sd);
        try {
            DFAgentDescription[] results = DFService.search(this, dfd);
            if (results == null || results.length == 0) {
                DFService.register(this, dfd);
                log("Registered in DF as " + this.type.getValue());
            }
        } catch (FIPAException e) {
            loge("Unable to register in DF");
        }
    }

    public void deregisterAgentDF() {
        try {
            DFService.deregister(this);
        } catch (FIPAException e) {
            loge("Unable to deregister from DF");
            doDelete();
        }
    }

    @Override
    public void doDelete() {
        deregisterAgentDF();
        super.doDelete();
        loge("Agent deleted");
    }

    public void log(String s) {
        System.out.println(System.currentTimeMillis() + ": " + getLocalName() + " - " + s);
    }

    public void loge(String s) {
        System.err.println(System.currentTimeMillis() + ": " + getLocalName() + " - " + s);
    }
}
```

Agentes de percepción (sensores)

- Disponemos de 4 agentes de percepción, siendo estos: lluvia, viento, humedad y temperatura.
- Cada uno de ellos tendrá un comportamiento de TickerBehaviour que mandará los datos correspondientes.

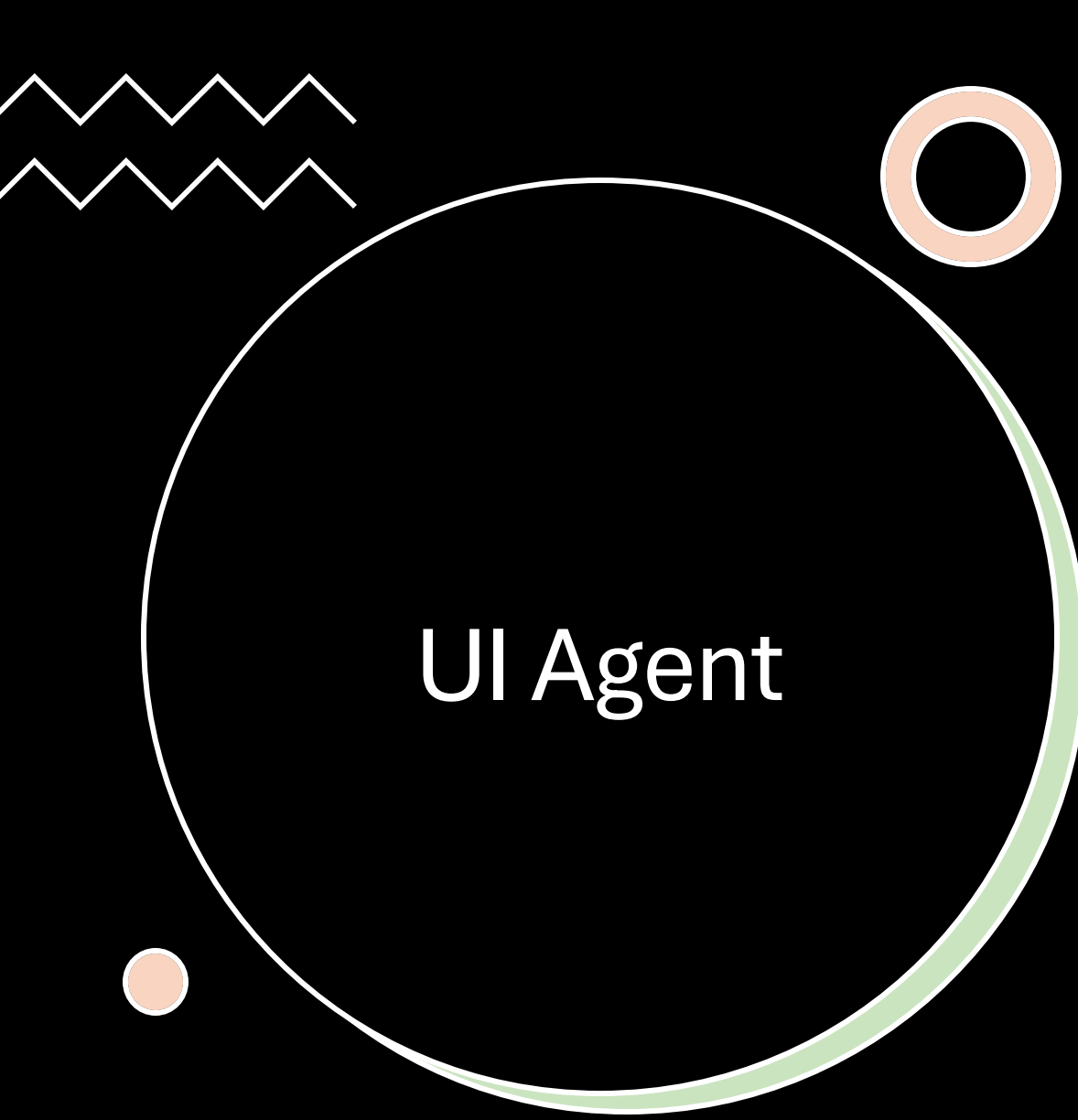
```
// Buscar al agente Decision en el DF cada cierto tiempo (una vez al inicio)
addBehaviour(new TickerBehaviour(this, 5000) { // cada 5 seg intenta encontrarlo
    protected void onTick() {
        if (decisionAgentAID == null) {
            DFAgentDescription[] results = getAgentsDF(AgentModel.DECISION);
            if (results.length > 0) {
                decisionAgentAID = results[0].getName();
                log("Decision agent found: " + decisionAgentAID.getName());
            }
        }
    }
});
```

Agente de procesamiento

```
// Comportamiento para recibir mensajes de sensores (Filtro por ontología)
addBehaviour(new CyclicBehaviour() {
    private MessageTemplate template = MessageTemplate.MatchOntology("weather-ontology");

    public void action() {
        ACLMessage msg = receive(template);
        if (msg != null) {
            processSensorData(msg);
        } else {
            block(); // Bloquea hasta que llegue un mensaje con esa ontología
        }
    }
});

private void processSensorData(ACLMessage msg) {
    String content = msg.getContent();
    log("Received: " + content);
    // Parsear contenido: "TEMPERATURE:21.5", "HUMIDITY:65.2", etc.
    String[] parts = content.split(":");
    if (parts.length == 2) {
        String type = parts[0];
        String value = parts[1];
        switch (type) {
            case "TEMPERATURE":
                latestData.put("temp", Double.parseDouble(value));
                break;
            case "HUMIDITY":
                latestData.put("hum", Double.parseDouble(value));
                break;
            case "WIND":
                latestData.put("wind", Double.parseDouble(value));
                break;
            case "RAIN":
                latestData.put("rain", value.equals("true") ? 1.0 : 0.0);
                break;
        }
        // Cada vez que se recibe un nuevo dato, evaluamos reglas
        evaluateAndDecide();
    }
}
```



UI Agent

- Tiene un comportamiento cíclico (CyclicBehaviour)
 - Para este apartado hemos usado la Inteligencia Artificial para ayudarnos a poder implementar la interfaz gráfica, mediante la herramienta Swing
- 

The background is a solid black field. A large, thin white circle is centered on the page. A thick, light green arc follows the bottom and right side of this circle. To the left of the circle, there are two horizontal wavy lines. Below them is a small, solid light orange circle. In the top right corner, there is a small double-lined circle, with the outer line being light orange and the inner line being white. In the bottom right corner, there is a rectangular grid of small white dots.

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